CSE 4074 Computer Networks

Fall 2021 Final Exam

11.02.2021 Friday

Duration: 40 Minutes for Part 1, Total: 105 Minutes.

Q1	Q2	Q3	Q4	Q5	SUM
/45	/10	/10	/20	/40	/100

Group-A

On my honor, I have neither given nor received any unauthorized assistance on this examination.

Name Surname:_____

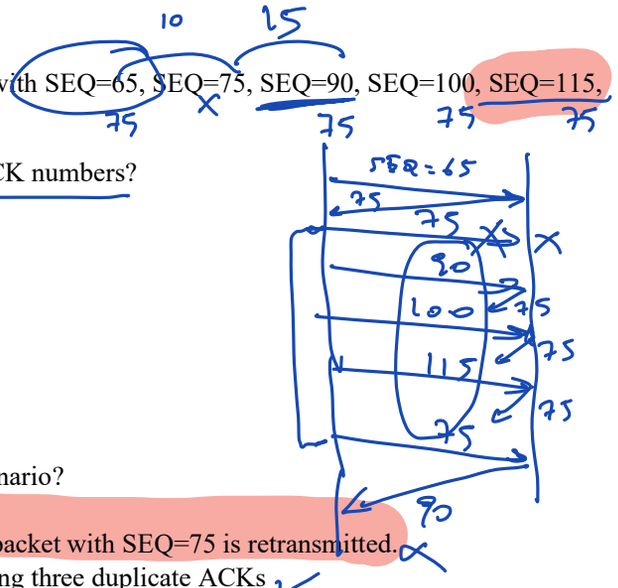
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Solution and Submission Notes

- All notes are closed. You are not permitted to refer to any source during the exam.
- Write your name, surname and student ID on the top of each exam solution page.
- Before starting to the exam, fill all the required areas in the test answer sheet provided to you. Do not forget to mark your group (A or B).
- The exam consists of two parts.
- In the first part, there are 21 multiple choice questions. You have to finish this part in 50 minutes, and mark your answers on the test answer sheet. After 50 minutes, test answer sheets will be collected.
- Each correct answer will bring +3 points, while each wrong answer will cause -0.75 points.
- The second part consists of two open-ended questions. Show all your work.

Q-1. (45 pts) Multiple choice questions.

Suppose that a TCP-sender sends only six segments with SEQ=65, SEQ=75, SEQ=90, SEQ=100, SEQ=115, SEQ=75. Answer questions 1-2.



1) Which one is a possible sequence of received ACK numbers?

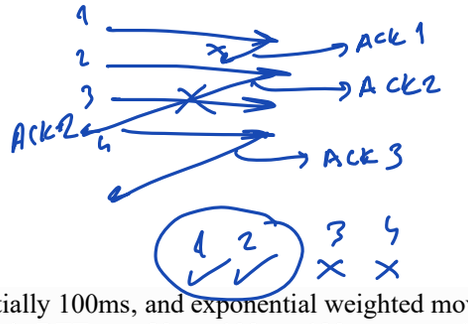
a. 65, 90, 100, 115, 75
b. 65, 75, 75, 75, 75
c. 75, 75, 75, 75, 90
d. 75, 90, 100, 115, 90
e. 75, 100, 115, 130, 90

2) Which of the following is false for the above scenario?

a. Definitely a time-out event is occurred and packet with SEQ=75 is retransmitted.
b. A fast-retransmit event occured after receiving three duplicate ACKs
c. The first segment is smaller than the second segment.
d. It is possible that the sender received just a single ACK for all six segments.
e. From the above information, we cannot know the length of the last segment.

3) Consider that a Go-Back-N sender sends four packets. The third packet is lost. Other packets are received by the receiver. The receiver sends ACKs and ACK of the first packet is also lost. Other ACKs are received by the sender. Which packets will be retransmitted by the sender?

a. Only the third packet
b. The first and the third packets
c. The third and the fourth packets
d. The first, the third and the fourth packets
e. All four packets



4) Suppose that EstimatedRTT for a TCP sender is initially 100ms, and exponential weighted moving average is used to update this value. If the next sample RTTs are 80ms, 100ms, 120ms respectively, which of the following is true?

a. The final value of the EstimatedRTT is less than 100ms.
b. The final value of the EstimatedRTT is equal to 100ms.
c. Sample RTT values do not change the value of EstimatedRTT.
d. It is suitable to set the timeout value for the next segment to 100ms.
e. None of the above.

5) Which of the following is the TCP field related to flow control?

a. Congestion control (cwnd)
b. Receive window (rwnd)
c. SYN
d. ACK number
e. C, E: Congestion notification.

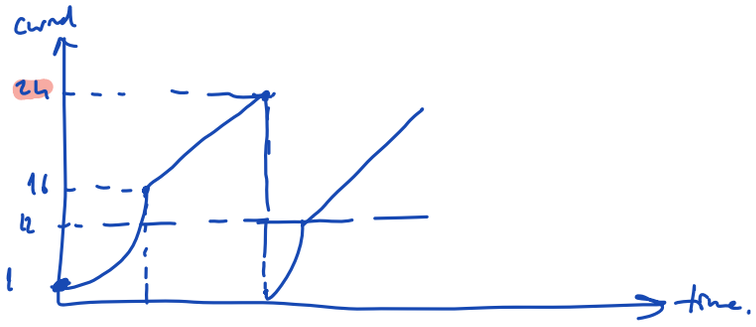
Timeout = EstimatedRTT + safety margin
4. DevRTT

100 80 100 120 100+

$$\left(\left(100\cdot\frac{7}{8}+90\cdot\frac{1}{8}\right)\cdot\frac{7}{8}+100\cdot\frac{1}{8}\right)\cdot\frac{7}{8}+120\cdot\frac{1}{8}$$

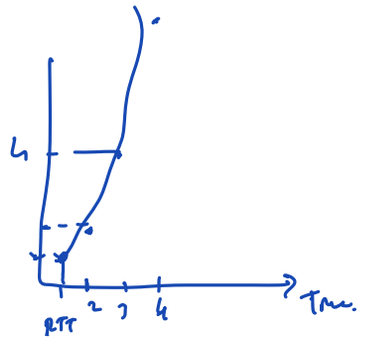
6) In a TCP, initially the threshold value (ssthresh) is 16 and a timeout event occurs when cwnd is 24. What would be the new value of ssthresh?

a. 8
b. 12
c. 16
d. 24
e. 40



7) What happens in the TCP slow start?

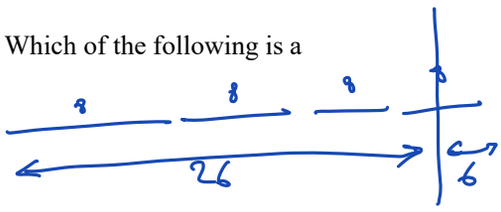
a. Cwnd is incremented by one for each transmission round.
b. Cwnd increases linearly, and then it starts increasing exponentially fast.
c. Cwnd almost doubles at each transmission round.
d. TCP reduces its sending rate.
e. A new but slow connection starts.



8) Suppose that IP address of host A is 192.167.3.235 and subnet mask is 26. Which of the following is a possible IP address of host B that is in the same subnet?

I. 192.167.2.235
II. 192.167.3.251
III. 192.167.3.100
IV. 192.167.3.256

a. Only I
b. Only II
c. II and III
d. II and IV
e. II, III and IV



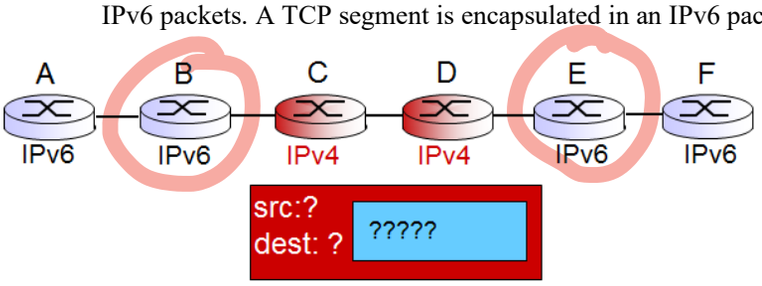
9) If data and the generator are both 1001, what would be the CRC bits?

a. 00
b. 000
c. 0000
d. 111
e. 1001

10) Which protocol is used by ping (to echo request and reply)

a. ICMP
b. DHCP
c. SNMP
d. NETCONF
e. Openflow

11) Consider a network where IPv6 routers and old IPv4 routers coexist. C and D are unable to process IPv6 packets. A TCP segment is encapsulated in an IPv6 packet and sent from A to E.



What would be the source and destination addresses and the payload of the IP packet sent from C to D?

a. A, F, TCP Segment
b. A, F, IP version 4 packet
c. C, D, IP version 6 packet
d. B, E, IP version 6 packet
e. B, E, IP version 4 packet

12) Which of the following information is typically not used in generalized forwarding?

a. Ingress port
b. IP source address
c. MAC source address
d. TCP destination port
e. HTTP host header

13) Which of the following is not a middlebox?

a. Network Address Translator
b. Firewall
c. Cache
d. Load balancer
e. DHCP server

14) In SDN, which of the following messages are sent from a router to controller, when a link fails?

a. features
b. configure
c. modify-state
d. port-status
e. flow-removed

15) Which of the following is not true?

a. If a lawful content is blocked by a router, although the network is idle, this can be considered as a violation of network neutrality
b. SNMP protocol uses UDP in the transport layer
c. IPv6 supports fragmentation
d. Within datacenters, Software Defined Networking (SDN) is widely used.
e. Virtual LANs may span multiple switches.

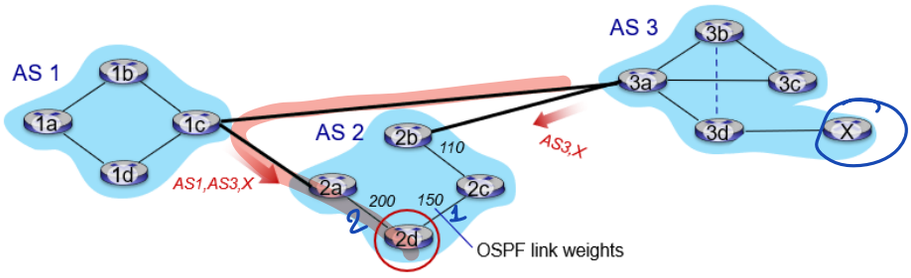
Q-2. (10 pts) a. Describe how TCP-Cubic increases utilization of channels.



b. A flow is considered TCP-friendly when it does not reduce the long-term throughput of any co-existent TCP flow more than another TCP flow on the same path would do under the same network conditions. Is UDP TCP-friendly? Why?

TCP has congestion control
UDP doesn't have

Q-3 (10 pts) Consider the following network. AS2 gateways receive two different path advertisements to subnet X. Suppose that 2d wants to send a packet to subnet X.



Answer the following questions:

a. Is it possible that 2d never learns a path via AS1 to subnet X? Why?

Yes. Due to policy restrictions

b. Suppose that 2d learns the both paths. What would be stored in its forwarding table for subnet X if SHORTEST AS-PATH route selection is used?

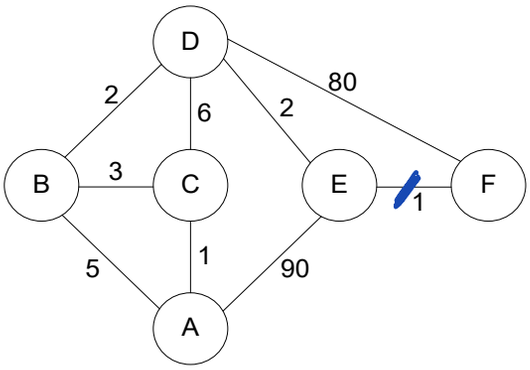
X - 1

F.T.

c. Suppose that 2d learns the both paths. Over which gateway would it send the packet if hot-potato routing is used? Why?

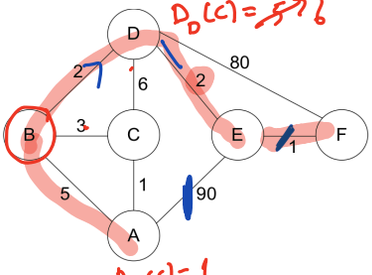
X - 2

Q-4. (20 pts) Consider the following network



(a – 8 pts) Show the operation of Dijkstra's (Link State) algorithm for computing the least cost path from **E to all destinations**. How does E set its forwarding table entries for all destinations if the traditional destination-based forwarding is used?

(b – 8 pts) Suppose that Distance Vector Routing is used and all the distance vectors are converged. Would removal of EF link cause to count-to-infinity problem? ~~What about~~ removal of AE link? Which one would converge slower? Show your work.



$D_D(C) = 5 \to 6$

$DV_D = (\ 6\ \ 2\ \ 5\ \ 0\ \ 2\ \ 3\ ) \to 7 \to 11 \to 15 \to \cdots \to 80$

$DV_E = (\ 8\ \ 4\ \ 7\ \ 2\ \ 0\ \ 1\ ) \to 5 \to 9 \to 13 \to \cdots \to 82$

Yes. There is C.I.P.

$D_A(C) = 1$

When AE is removed, non of the DVs would change. Therefore there is no C.I.P.

(c – 4 pts) Now suppose that we use Software Defined Networking (SDN) approach (Openflow) and we want all the packets from A to E to follow the path A-B-D-E instead of the shortest path. How should we set match and action fields in ~~Node~~ D?

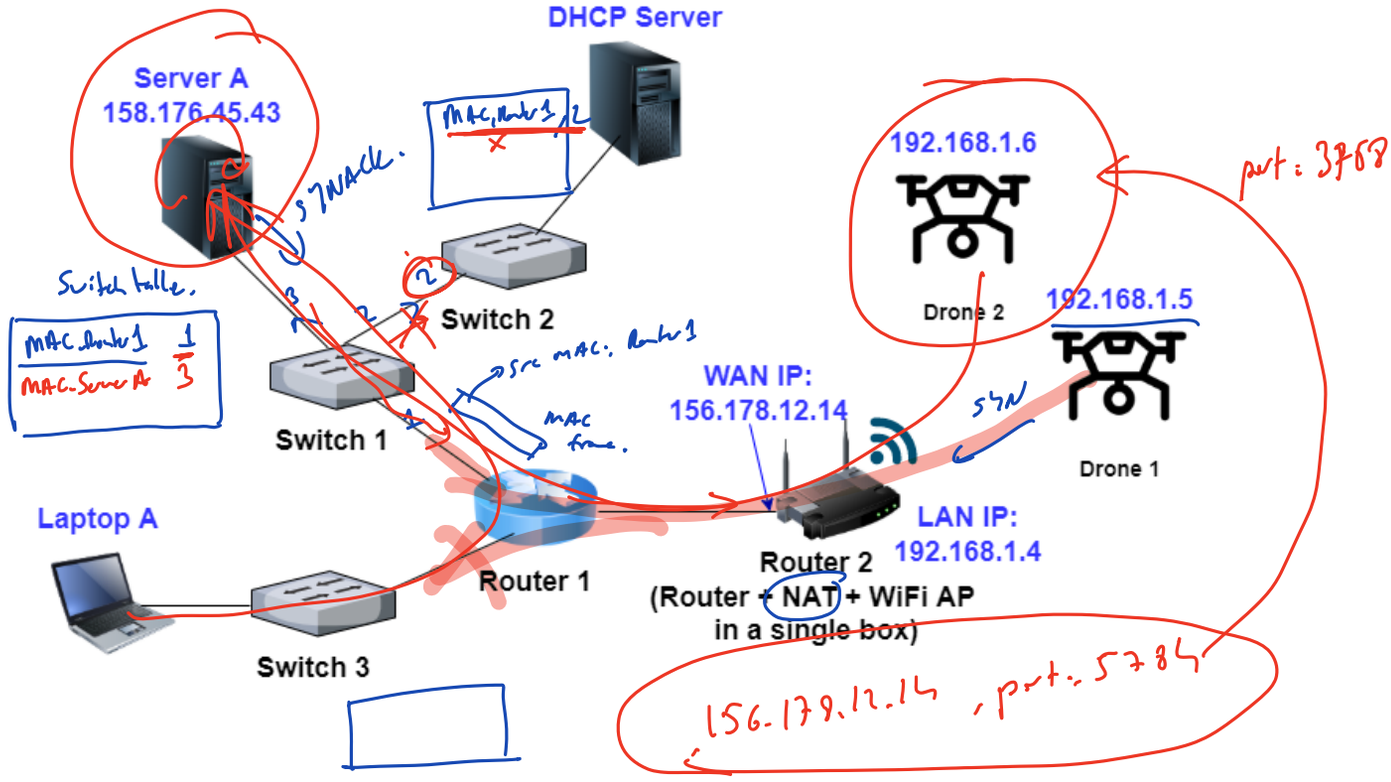
Shortest Path : A – C – B – D – E.

A: match: src: A, dest: E output: (A, b)

B: inp: AB, src: A, dst: E output: (B, D)

D: inp: BD, src: A (IP), dst E (IP) output (D, E)

Q-5. (40 pts) Consider the following network. Drones are connected to Router 2. Answer the following questions.



a. (8 pts) Let us say that all switch tables are initially empty. Drone 1 sends a SYN message to Server A. Describe what actions are taken in (i) Router 2 (NAT); (ii) Switch 1; (iii) Switch 2; and (iv) Switch 3. (give any information that is stored in these nodes as well)

Router 2 : NAT table :
192.168.1.5, port# ⇒ newport#, WAN-IP.
Router 2 modifies src IP addr as WAN-IP
src port # as new port #.

b. (7 pts) Server A sends back a SYNACK message to Drone 1. Describe what actions are taken in (i) Router 2 (NAT); (ii) Switch 1; (iii) Switch 2; and (iv) Switch 3. (give any information that is stored in these nodes as well)

Router 2 checks the NAT table
and modifies destination IP addr as 192.168.1.5
dest port as port of the Drone 1 process.

c. (4 pts) Let say that we attach Laptop A to switch 3. Laptop A requires to communicate with DHCP server to get a new IP address. How can it send a request to the DHCP server without knowing any IP address?

Broadcasting.
dest IP: 255.255.255.255.
dest MAC: FF.FF.FF.FF.FF.FF.

d. (7 pts) After connecting, Laptop A wants to send a request to Server A. Does it need the IP address of Router 1? Why or why not? If it needs, how it learns it?

dst MAC: Router 1's MAC.

ARP.

e. (7 pts) Suppose that Drone 1 and Drone 2 uses Carrier Sense Multiple Access (CSMA) while sending a frame to Router 2. Is there any effect of the distance between drones on the collision probability? Why? What would be an advantage of using CSMA rather than a collision-free protocol like FDMA?



f. (7 pts) Describe the problem encountered when Laptop A wants to establish a connection with Drone 2. Describe a reasonable solution for this problem, by only using the hosts and network entities given in the figure.